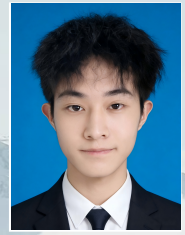


Yongxue Xu (Jerry)

VIDEO INTELLIGENCE / GENERATIVE WORLD MODELS / 4D REASONING

xuwx85@mail2.sysu.edu.cn • jerrysnow.me • github.com/lamJerryXu

Google Scholar • LinkedIn • Sun Yat-sen University



SUN YAT-SEN UNIVERSITY

Profile

01

Undergraduate researcher at Sun Yat-sen University working on video understanding and generation, multimodal foundation models, world models, and 4D scene understanding. I am interested in systems that connect perception, generation, and interaction, and I am currently seeking industry research internships in these areas.

Education & Toolkit

02

Sun Yat-sen University

B.Eng. in Intelligent Science and Technology

GPA 3.91 / 4.0 • Major GPA 3.94 / 4.0

Project 985 / 211 / Double First-Class

SEP 2023 -- JUN 2027

RESEARCH

Video understanding/generation, multimodal foundation models, world models, 4D scenes, WAM.

TOOLS & LANGUAGES

Python, C/C++, MATLAB, PyTorch; Chinese (native), English (CET-6).

Research Experience

03

01 LMM-Track4D

CO-FIRST / NEURIPS 2026

Built a three-domain, multi-view temporal benchmark and a trajectory-grounded framework for object-level 4D reasoning with streaming state tokens and 3D trajectory decoding.

CONTRIBUTION Led the technical roadmap, model design, and main experiments; contributed to benchmark construction.

02 POSReasoner

CO-FIRST / PREPRINT

Developed a plug-and-play framework that retains, verifies, and reactivates persistent object states through a sparse state-observation graph while keeping the host model frozen.

CONTRIBUTION Led the roadmap, architecture, implementation, experiments, and manuscript.

03 EviGDA

CO-FIRST / PREPRINT

Combined entropy-aligned graph-aware and graph-free experts through post-training fusion, reaching state-of-the-art results on 10 datasets and 16 transfer tasks without learned routing or target pseudo-labels.

CONTRIBUTION Designed the core method and led implementation, evaluation, and manuscript development.

04 DRCIM-ML

FIRST / TNSE REVIEW

Created a diffusion-aware role-guided evolutionary framework with cross-layer alignment, role constraints, dynamic robustness evaluation, and multi-task search.

CONTRIBUTION Led the roadmap, core experiments, patent filing, and manuscript; one patent pending.

05 Urban Critical Node Identification

THIRD / BEST PAPER

Evaluated GNN and DQN approaches for critical-node discovery on real urban road networks.

CONTRIBUTION Contributed dataset construction, baseline reproduction, and experimental visualization.

Research Index

YONGXUE XU / 2025--2026

Selected Publications

04

- 01

LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue
Chaoyue Li*, **Yongxue Xu***, Jie Feng*, and Jiayu Ding†

2026

Submitted to NeurIPS 2026

- 02

Learning to Reason with Persistent Object States for Video Instance Segmentation
Yongxue Xu*, Boxue Yang*, Ziqian Liu, Shaoqiu Zhang, and Linfeng Zhang†

2026

Preprint

- 03

Graph Domain Adaptation Does Not End with Representation Learning
Yongxue Xu*, Ziqian Liu*, Enze Zhang, and Maolin Wang†

2026

Preprint

- 04

Solving the Robust Influence Maximization Problem in Competitive Multilayer Networks via a Diffusion-Aware Role-Guided Evolutionary Approach
Yongxue Xu, Ziqian Liu, Yongqing Huang, and Shuai Wang†

2026

Under review at IEEE Transactions on Network Science and Engineering

- 05

Identifying Critical Nodes with Deep Learning and Reinforcement Learning: A Case Study on Urban Road Networks
Zhonghan Liu, Wenshuo Liu, **Yongxue Xu**, Ziqian Liu, Yichen Liu, and Shuai Wang†

2025

GBCESC 2025 / Best Paper Award

* Equal contribution. † Corresponding author.

Research Initiatives & Collaborations

05

I collaborate with researchers at **HKUST**, the **EPIC Lab at SJTU SAI**, and **Westlake University** on video generation, world-action modeling, and multimodal spatiotemporal understanding. I also co-organize **InkMind.AI**, a small group exploring multimodal understanding and generation. Through vibe coding, we develop RSI and auto-research workflows that help AI frame questions, build systems, evaluate results, and iterate with greater precision and efficiency. We see AI-builds-AI as a promising next step.

VIDEO GENERATION WORLD-ACTION MODELS MULTIMODAL REASONING AI BUILDS AI

Awards & Competitions

06

- 01

Chinese Mathematics Competition (CMC), Guangdong Region – Second Prize

Dec. 2024
- 02

Mathematical Contest in Modeling (MCM), USA – Honorable Mention

Feb. 2025
- 03

National Undergraduate Smart Car Competition, South China Region – Second Prize

Aug. 2025

Beyond Research

07

Table tennis, music, and nature travel.